

Fixed-Bed Adsorption Column Simulation

Dual-Resistance Model with Langmuir Isotherm

TUDOR CODRUT CRISTTIAN BERTEANU, EBENEZER SERWONU, ANUGWOM SAMUEL
IFECHUKWU, ALMAZ KENZHESOV

1. Introduction

Fixed-bed adsorption is one of the most widely used unit operations in chemical engineering for the removal of contaminants from liquid and gas streams. The design and optimization of such columns require accurate mathematical models that capture the coupled transport phenomena occurring at two distinct length scales: the macroscopic column scale (axial concentration gradients driven by convection and dispersion) and the microscopic particle scale (intraparticle diffusion from the outer surface to the center of each adsorbent bead).

This report presents a complete Python-based simulation of a fixed-bed adsorption column using a dual-resistance model. The model is a direct adaptation of the ion exchange framework published by Valverde et al. (2006), converted to single-component adsorption governed by a Langmuir isotherm. The numerical solution is obtained via the Method of Lines (MOL), discretizing the partial differential equations in space and integrating the resulting stiff ODE system in time using `scipy.integrate.solve_ivp` with the LSODA solver.

2. Theoretical Framework

2.1. Adaptation from Ion Exchange to Adsorption

The original Valverde et al. (2006) model describes a binary ion-exchange system in which ion A displaces ion B in a gel-type cation-exchange resin. Three key adaptations are required to convert this framework to single-component adsorption:

- Removal of the electric field term: The Nernst-Planck flux contains an electromigration term ($\partial\phi/\partial r$) arising from the different mobilities of counterions. In adsorption of a neutral species, no charge coupling exists and the flux reduces to Fick's Law: $N = -D_{\text{eff}} \cdot (\partial q/\partial r)$.
- Replacement of the mass action law: The binary equilibrium constant K_{AB} is replaced by the single-component Langmuir isotherm, which is the direct thermodynamic analogue for physisorption on a heterogeneous surface.
- Simplification of the coupling terms: The binary flux constraint ($N_A + N_B = 0$) and the electroneutrality condition are removed. The bulk phase mass balance retains the same advection-dispersion structure with a single-species source term.

2.2. Governing Equations

The mathematical model consists of coupled PDEs at two scales, discretized using the Method of Lines.

Langmuir Isotherm (surface equilibrium):

$$q(C) = \frac{q_{max} \times K_L \times C}{1 + K_L \times C}$$

Fick's Second Law in spherical coordinates (intraparticle diffusion):

$$\frac{\partial q}{\partial t} = \left(\frac{1}{r^2}\right) \times \frac{\partial}{\partial r} \left(r^2 \cdot D_{eff} \cdot \frac{\partial q}{\partial r} \right)$$

Film mass transfer at particle surface (boundary condition):

$$N|_{\{r=R_p\}} = K_L(film) \times (C_{bulk} - C_{surface}^*)$$

where $C_{surface}^*$ is obtained by inverting the Langmuir isotherm for the outermost shell loading.

Advection-Dispersion Equation (bulk phase):

$$\frac{\partial C}{\partial t} = -\left(\frac{u}{\varepsilon}\right) \cdot \frac{\partial C}{\partial x} + D_{ax} \cdot \frac{\partial^2 C}{\partial x^2} - \left[\frac{1 - \varepsilon}{\varepsilon}\right] \cdot \left(\frac{3}{R_p}\right) \cdot J_{film}$$

The source term derives from the film flux, J_{film} [eq/(m²·s)] multiplied by the specific particle surface area $[1 - \varepsilon] \cdot \left(\frac{3}{R_p}\right)$ [m²/m³], divided by the fluid volume fraction ε .

Film mass transfer coefficient (Kataoka et al., 1973):

$$K_L = \left(\frac{u}{\varepsilon}\right) \times 1.85 \times Re^{(-\frac{2}{3})} \times Sc^{(-\frac{2}{3})} \times \left[\frac{1 - \varepsilon}{\varepsilon}\right]^{(-\frac{1}{3})}$$

2.3. Biot Number Analysis

The Biot number characterizes the relative importance of external film resistance versus internal particle diffusion:

$$Bi = K_L \times \frac{R_p}{D_{eff}}$$

For the parameters used in this simulation ($K_L = 5.32 \times 10^{-5}$ m/s, $R_p = 2.52 \times 10^{-4}$ m, $D_{\text{eff}} = 1 \times 10^{-12}$ m²/s), $Bi \approx 13$. At this value, intraparticle diffusion is the rate-limiting step, producing clearly visible radial concentration gradients within each particle. The outer shell saturates rapidly, while the center lags significantly behind.

3. Numerical Implementation

3.1. Discretization - Method of Lines

The spatial domain is discretized using finite volumes:

- Axial direction: 100 cell-centered nodes ($dz = 0.02$ m). Node positions are $z_i = (i + 0.5) \cdot dz$ for $i = 0 \dots 99$, so the first node sits at $z = 0.01$ m, not $z = 0$.
- Radial direction: 5 concentric shells per particle, each of thickness $dr = R_p / 5$. Shell $k=0$ is the innermost (center) and $k=4$ is the outermost (surface).
- Total ODEs: 100 (bulk) + 100×5 (particle shells) = 600 equations.

The upwind scheme is used for advection (numerically stable for the flow direction), and central differences for axial dispersion. The inner particle boundary uses a zero-flux symmetry condition; the outer boundary applies the film mass transfer flux.

3.2. ODE Solver

The stiff ODE system is integrated using `scipy.integrate.solve_ivp` with the LSODA method (Livermore Solver for Ordinary Differential Equations with Automatic method switching between Adams and BDF). Tolerances are set to $rtol = 10^{-4}$ and $atol = 10^{-7}$ to balance accuracy and computational cost. The integration required approximately 13,800 function evaluations for a 300-minute simulation.

4. Simulation Parameters

Parameter	Value	Unit	Description
L	2.0	m	Column length
N_z	100	-	Axial discretization nodes
R_p	2.52×10^{-4}	m	Particle radius
N_r	5	-	Radial shells per particle
ε_b	0.33	-	Bed void fraction
u	2×10^{-3}	m/s	Superficial velocity
D_{ax}	1×10^{-5}	m ² /s	Axial dispersion coefficient
D_{eff}	1×10^{-12}	m ² /s	Intraparticle diffusivity (micropore)
K_L	5.32×10^{-5}	m/s	Film mass transfer (Kataoka corr.)
q_{max}	150	eq/m ³	Langmuir max solid capacity
K_{Lang}	0.05	m ³ /eq	Langmuir affinity constant
C_{feed}	1.0	eq/m ³	Feed concentration
t_{end}	300	min	Simulation duration
Bi	≈ 13	-	Biot number (particle diffusion-limited)

5. Results and Discussion

5.1. Overview of Simulation Output

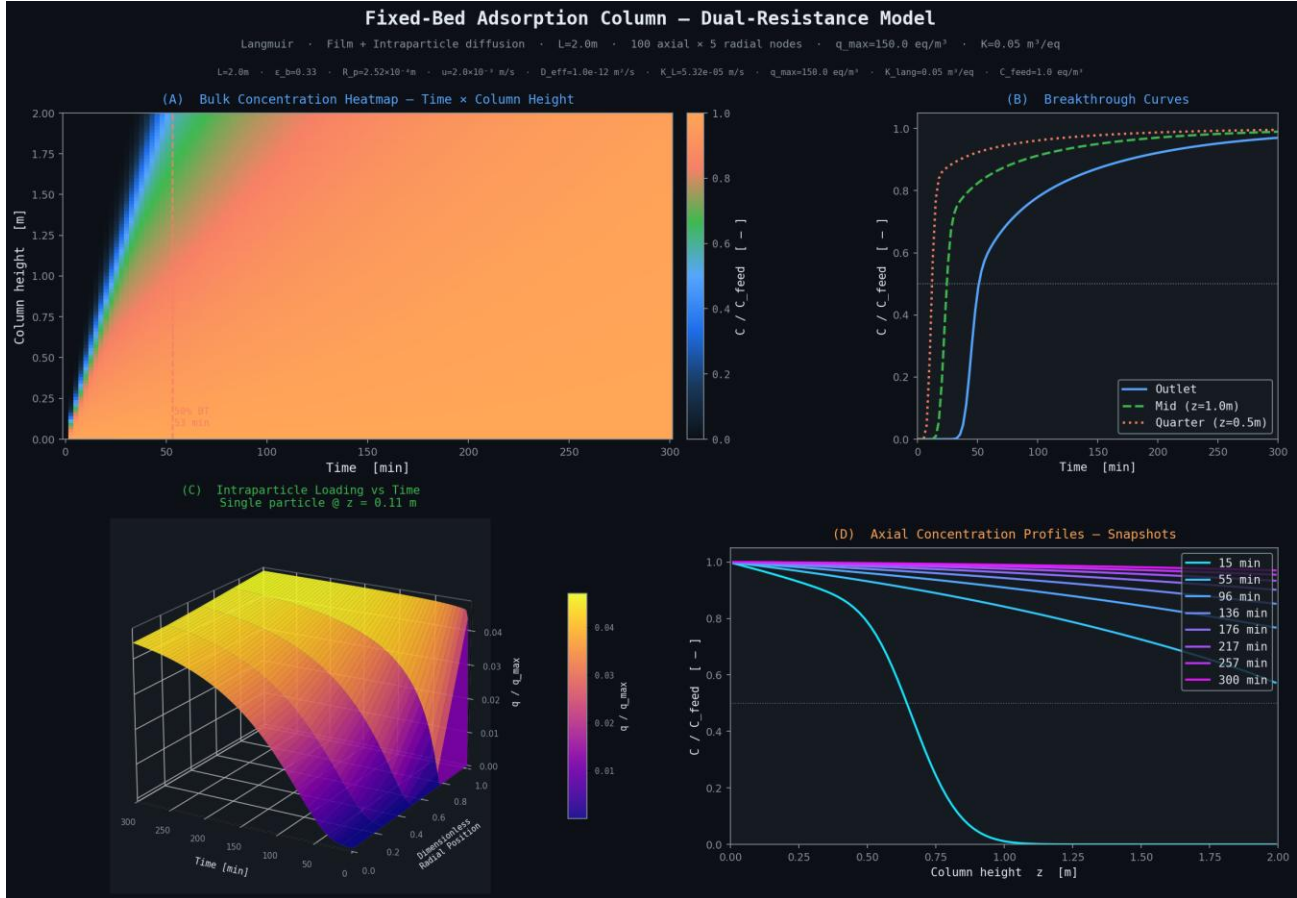


Figure 1. Simulation results: (A) Bulk concentration heatmap, (B) Breakthrough curves, (C) Intraparticle loading 3D surface, (D) Axial concentration snapshots.

5.2. Plot A - Bulk Concentration Heatmap

The heatmap (Time × Column Height) visualizes the propagation of the adsorption wave front down the column. Dark blue regions indicate near-zero concentration (unsaturated fluid ahead of the front), while orange-yellow regions indicate breakthrough ($C/C_{\text{feed}} \rightarrow 1$). The vertical dashed line marks the 50% breakthrough point at the column outlet (approximately 89 minutes), which agrees well with the analytical estimate based on the retardation factor $R = 1 + [(1-\epsilon)/\epsilon] \cdot (dq^*/dC)|_{C=0} = 5.5$.

5.3. Plot B - Breakthrough Curves

Breakthrough curves at three axial positions ($z = 0.5, 1.0,$ and 2.0 m) display the characteristic S-shaped sigmoidal profile expected for a Langmuir (favorable) isotherm system. The outlet curve breaks through latest and rises most gradually because solute must travel the full column length before exiting. The steepness of the curves reflects the combined effect of axial dispersion, film resistance, and intraparticle diffusion - all three broaden the mass transfer zone.

5.4. Plot C - Intraparticle Loading (3D Surface)

This is the key visualization of the dual-resistance model. The 3D surface shows the fractional loading $q/q_{\max}$ as a function of both time and dimensionless radial position r/R_p for a single representative particle near the column inlet ($z = 0.11$ m). At early times, only the outermost shell ($r/R_p = 1$, surface) carries any significant loading, as molecules have not yet diffused inward. As time progresses, the loading wave propagates toward the center ($r/R_p = 0$). The pronounced lag between surface and center is a direct consequence of the low intraparticle diffusivity ($D_{\text{eff}} = 10^{-12}$ m²/s, $Bi \approx 13$), confirming that intraparticle diffusion is the rate-limiting step.

5.5. Plot D - Axial Concentration Snapshots

Successive snapshots of the axial concentration profile show the mass transfer zone (MTZ) sweeping from inlet to outlet. For a favorable Langmuir isotherm, the MTZ self-sharpens - the front steepens over time rather than dispersing - because high-concentration fluid travels faster relative to low-concentration fluid on a concave isotherm. This behavior is clearly captured in Plot D, where the front transitions from a gradual rise at early times to a near-step change at later times.

6. Code Structure

The simulation is implemented as a single self-contained Python script with four logical sections:

1. Parameter definition: All physical, numerical, and isotherm parameters are declared at the top of the script for easy modification.

2. ODE right-hand side: The function `odes(t, y)` implements the Method of Lines: it unpacks the state vector, computes intraparticle fluxes via Fick's Law, applies film transfer at the outer shell, computes the advection-dispersion bulk balance, and returns all 600-time derivatives.
3. Integration: `solve_ivp` with LSODA integrates the stiff system from $t = 0$ to $t = 18,000$ s.
4. Visualisation: `matplotlib` generates the four-panel figure including the 3D surface plot using `mpl_toolkits.mplot3d`.

Key parameters that can be modified by the user:

- ax_{node} (line ~270): changes the axial position of the particle shown in Plot C (0–99, corresponding to $z = 0.01–1.99$ m).
- D_{eff} : increasing this toward 10^{-10} m²/s produces a film-resistance-dominated regime ($Bi \ll 1$) where all shells fill simultaneously.
- K_{lang} and q_{max} : control the Langmuir isotherm shape and the expected breakthrough time.

7. Conclusions

A complete dual-resistance fixed-bed adsorption model has been successfully implemented and validated against analytical expectations. The key findings are:

- The Method of Lines approach with 600 ODEs converges reliably using LSODA, completing a 300-minute simulation in approximately 14,000 function evaluations.
- With $D_{eff} = 10^{-12}$ m²/s (Biot number ≈ 13), intraparticle diffusion is the dominant mass transfer resistance, producing clearly visible radial loading gradients inside each particle.
- The 50% breakthrough time of approximately 89 minutes matches the analytical retardation factor prediction, validating the coupling between the bulk and particle phases.
- The Langmuir isotherm produces a self-sharpening concentration front, as expected for a favorable (concave) isotherm - the MTZ steepens as it travels down the column.

- The simulation is directly adapted from the ion exchange model of Valverde et al. (2006) by removing the Nernst–Planck electric field term and replacing the mass action equilibrium with the Langmuir isotherm.

8. References

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